

Mia Kiesman

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Lumafield, Systems Engineer II

April 2025 – Present

- Led embedded development of a metrology-grade CT scanner, performing design, software development, prototype assembly, and motion tuning to micron level accuracy. Enabled mechanical & optical validation within weeks and finalized production-ready designs in less than three months.
- Developed CT calibration & verification programs to increase accuracy across hundreds of scanners.
- Wrote Python microservices for factory robotics systems which enabled failure tracing and fault recovery, along with performing pre-production electrical and embedded troubleshooting. This ensured the first unit shipped on time while meeting reliability goals within 2 months of onboarding.

Generac Clean Energy, Firmware Engineer

May 2024 – April 2025

- Wrote automated UL1998 HiL testing suites for TI C2000 based control modules
- Maintained CI/CD pipeline which performed Linux and embedded target compilation, static analysis, unit tests, and HiL/SiL verification using PyTest and PLECS, enabling better code quality across team
- Implemented bare-metal C isoSPI BMS driver, including safety critical controls, and integrated it into pre-existing codebase, enabling robust hotswap of heterogeneous battery chemistries into field units

MIT Lincoln Labs, Energy Systems Group, Research Intern

June 2023 – August 2023

- Developed automated IP network infrastructure analysis tools for use in power grid settings
- Integrated industrial internet protocols into Hardware in the Loop (HiL) testing infrastructure to enable faster deployment of new devices and richer data collection for device testing
- Evaluated Software Defined Networking solutions for their suitability in industrial applications

Generac Clean Energy, EE Integration Engineering Intern

May 2022 – August 2022

- Wrote and deployed real-time production code for residential solar/battery microgrid installations
- Worked on an agile development team maintaining a large multi-core embedded C codebase

Spin Analytical, Electrical Engineer

June 2017 – August 2021

- Designed, built and delivered an automated, microfluidics platform to incubate, measure and purify radioactive solutions for a top pharmaceutical manufacturer's pilot program to automate drug synthesis.
- Integrated hardware including a sensor array, robotic gantry and pipet system.
- Planned and wrote user interface, component drivers, analysis protocol and bench test modules.
- Created hardware and software for optical centrifuge rotor calibration deployed in over 50 research labs. This enabled faster and more accurate charge measurement of protein solutions.

Permit Log, Developer

February 2017 – 2019

- Won first place in the Maine Apps Challenge and was awarded a \$6000 scholarship
- Supported over 3000 active Android users over multiple app versions

Education

Columbia University, Electrical Engineering, 3.64 GPA

Class of 2024

Embedded AI/ML, FPGA SoC design, Power Electronics, Vision in Robotics, Communication Circuits

Marshwood High School, South Berwick, ME

Class of 2019

Skills

KiCAD, LTSPICE, EAGLE, RP2040, Arduino, ESP32, STM32, C2000, RP2040, C, C++/Qt, Java, Python, Rust, TensorFlow, Pascal, Mathematica, Linux/Windows/macOS development, Analog/Digital PCB design, Final Cut Pro, Adobe Premiere

Independent and University Projects

ModMatic - Senior Design Capstone

September 2023 – May 2024

Worked on a team of 3 to create an RP2040 based musical instrument which enables the player to create and play arbitrary microtonal scales. My area of responsibility was hardware design and layout of the two interconnected PCBs, and writing the ARM C++ firmware for each board

XTrip

September 2020 – August 2021

Created a realtime internet audio bridge and serial interface to restore service to a defunct 90s multiplayer modem-based gaming network

CondWand - Embedded AI Final Project

March 2023 - May 2023

Created a prototype wearable bracelet to assist in training of band conductors. Used a 6 axis IMU to collect training data and provide realtime feedback to the user.

MOP3

January 2023 - April 2023

Open source, Rust based email daemon created to allow retro computers to browse the Mastodon social network. Supports POP3, IMAP, images, and embedded images/HTML.

!!Con, Speaker

May 2021

- Presented talk on the use of the Teensy Microcontroller to play real instruments in rhythm games.

Small File Film Festival, Filmmaker

August 2020

- Won *Best Cross-Platform Work* for entry "New Beginnings", created for the Gameboy Advance

2600, Writer

April 2020

- Wrote an article on Internet preservation and archival for hacking/technology magazine *2600*